Theoretical and Applied Mechanics
F.I. Niordson and N. Olhoff
Elsevier Science Publishers B.V. (North-Holland)

HYDRODYNAMIC SYNTHESIS OF MARINE STRUCTURES

Professor J. R. Paulling

Department of Naval Architecture and Offshore Engineering
University of California
Berkeley, California 94720 USA

The diverse and sometimes irregular geometry of offshore structures, together with their exposure to a severe dynamic marine environment, results in a system of hydrodynamic loads whose origins lie in several different fluid phenomena. The fundamental reasons for computing the hydrodynamic forces are twofold: first, they are needed for computing the motions of the floating structure in waves and second, they form an important part of the structural load system. The characteristics of structure geometry and motion environment which influence the above phenomena will be outlined. The various components of loads and their interaction will be discussed and we examine the techniques currently in use for design analysis.

INTRODUCTION

The term synthesis is defined as "the putting together of parts to form the whole" and is quite appropriately applied to the process of determining the hydrodynamic forces on marine structures of the type used in offshore oil drilling and other similar operations. In contrast to ships, which usually have a hydrodynamically smooth and regular surface, offshore platforms are constructed in the form of an irregular three-dimensional assembly of pontoons, columns and braces, often of cylindrical or near cylindrical configuration.

These offshore platforms may be subdivided into two very general categories of fixed and floating types, respectively. Fixed platforms are attached to the seafloor by a combination of their weight and by piling driven into the sea bed. Floating platforms may be held in place by means of an array of chains and cables with anchors at the sea bed, or they may be dynamically positioned, i.e., held in place by means of computer controlled thrustors which are activated in such a way as to maintain the platform in position over a fixed location on the bottom. Some platforms are supported by raising themselves above the sea surface on legs which extend to the sea floor. For transportation, such jackup platforms are lowered until they float on the main hull, which resembles a barge, and the legs are raised into an elevated position above the platform. A recent innovation in platform design is the tension leg platform or TLP. This is very similar to the moored platform with the exception that the buoyancy exceeds the weight and vertical force equilibrium is maintained by means of taut vertical mooring members extending from the platform to the sea floor.

In the present paper, we shall concern ourselves principally with the floating platforms. Fixed platforms of the pile-supported type are subject to many of the same wave-structure interactions with the exception of those due to the platform's own motion. We shall discuss some of the important considerations involved in making estimates of the hydrodynamic forces on such structures and deducing their response. Particular attention will be paid to those areas in which the designer and engineer is forced to make simplifications and approximations as a result of deficiencies in the relevant hydrodynamic theories.

The most common geometric configuration for both the semisubmersible and the TLP consists of a space-frame arrangement of tubular members of circular or oval cross-section. Platforms of somewhat simplified geometry but having proportions typical of modern offshore oil field drilling and production applications are shown in Figures 1 and 2. The twin hull arrangement has become, by far, the most widely used arrangement for the semisubmersible, dictated largely by considerations of low resistance to forward motion when in transit. The first full scale working TLP has recently been installed in the North Sea and operational experience with the concept is yet to be obtained. The TLP shown in Figure 2 is similar to several conceptual designs which have been developed by various oil companies. Since most of these have been intended for oil production rather than exploratory drilling, mobility has not influenced the arrangement to the same extent that it has for the semisubmersible. Considerations of optimum mooring tension response have resulted in somewhat larger vertical members and smaller horizontal members as compared to the semisubmersible.

Some important characteristic of both of these platforms are the following:

Much of the immersed volume is deeply submerged and the waterplane area is relatively small in relation to the total volume.

Most of the tubular members are relatively small in cross section compared to their own length and to the length of important wave components.

The spacing between members is large compared to the member cross section.

The first of the above characteristics gives the platform its "wave transparency" properties. The second and third may be utilized in deriving an analysis of the wave-induced forces and resultant motions which, although greatly simplified, nevertheless is quite accurate and informative in developing understanding of the principles upon which the two types of platforms may be designed.

GENERAL CONSIDERATIONS

In solving for the hydrodynamic loads and resulting motion of the platform, it is usually acceptable to treat the platform as a rigid body since the elastic deformations are small compared to the wave and body motions which give rise to the forces. The system of external loads acting on the body fall into several different categories, including:

(1) Environmental forces caused by waves, current and wind.

(2) Forces due to the platform motion.

(3) Restraint forces exerted by the mooring or positioning system.

(4) Forces associated with the platform operations such as heavy lifts, riser tension, drill pipe forces, and contact with attendant service ships.

The initial objective in the problem solution is to find the time history of the relationship between two coordinate systems, one fixed in the platform and the other fixed in space, and these are illustrated in Figure 3. After solving for the motions of the platform, we may add to the above system of external forces the internal inertia loads of the distributed mass of the platform and its contents to obtain a self-equilibrating system of structural loads. These loads may be used as input, for example, to a finite-element structural analysis.

The position of the platform is defined at any time by means of a set of three displacements of its origin along the space axes and a vector of three Euler angles representing rotations about three successive orientations of the body axes. At any instant of time, Newton's second law may be stated:

$$\underline{F} = \frac{d}{dt}(\underline{m}\,\underline{v})$$
$$\underline{M} = \frac{d}{dt}(\underline{I}\,\underline{\omega}) = \underline{I}\,\dot{\underline{\omega}} + \underline{\omega} \times (\underline{I}\,\underline{\omega}) \tag{1}$$

Here $\underline{m}$, $\underline{I}$ = the mass and moment of inertia matrices representing the physical mass of the platform itself.

$\underline{F}$, $\underline{M}$ = vectors of external forces and moments.

$\underline{v}$ = translatory velocity of the C.G. in global coordinate system.

$\underline{\omega}$ = angular velocity vector in the body coordinate system.

The equations of motion are nonlinear both as a result of the product of angular velocities in the second of (1) and through nonlinear relationships between the external forces and the platform motions, waves, currents and other influences. The force and moment terms include all of the external forces listed above, and much of the effort in obtaining the solution to (1) is devoted to the determination of these forces. In the design of platforms, several different approximations are used in order to obtain estimates of these forces. The method used in a given situation depends upon the geometry of the structure, the severity of the environment being simulated and the computational means at the disposal of the user.

In the case of small waves and small platform motions, two important simplifications are possible. First, the equations may be linearized, and second, the external hydrodynamic forces may be replaced by a simple sum of four terms proportional respectively to the absolute acceleration, the absolute velocity, the displacement from the mean position and a term independent of the body motion. The result is expressed as follows:

$$\underline{m}\,\ddot{\underline{x}} = -\underline{m}'\,\ddot{\underline{x}} - \underline{b}\,\dot{\underline{x}} - \underline{c}\,\underline{x} + \underline{F}_w(t) \tag{2}$$

Here $\underline{m}$ = composite matrix of mass and moment of inertia.

$\underline{x}$ = composite vector of displacements of c.g. and Euler angles.

$\underline{m}'$ = added mass matrix for structure.

$\underline{b}$ = damping matrix

$\underline{c}$ = restoring force matrix including hydrostatic and mooring effects.

$\underline{F}_w(t)$ = excitation force caused by external effects such as waves.

In general, the coefficients $\underline{m}'$, $\underline{b}$ and $\underline{F}_w$ depend upon the frequency of the incoming waves, ω . Consequently, equation (2) cannot be integrated directly for any arbitrary $\underline{F}_w(t)$. In the case of random excitation such as that due to realistic ocean waves, we may solve a separate equation (2) for each of the composite waves into which the random seaway may be decomposed and then superimpose these individual regular wave solutions. Thus, (2) reduces to an algebraic equation for the complex motion amplitude in the frequency domain. If we wish to deal with the complete nonlinear equation system (1) by a direct numerical integration scheme, it is necessary first to transform these frequency-dependent terms into an impulse response function in the time domain by evaluating their Fourier transform. A convolution integral is then employed to express the force in the time domain.

Much insight and understanding of the respone characteristics of ocean platforms may be gained by examining the solutions to (2). There are, however, several important phenomena and effects which are lost in the process of linearization and which may only be obtained by solving the exact equations, (1).

The response of the platform to waves, as in the case of any oscillating system, depends upon two quantities, the exciting force and the resonant response of the system to unit excitation. Both of these quantities are found to vary with the frequency of the incident waves and, consequently, the total response characteristics of the structure to waves depends upon the relationship between the two functions.

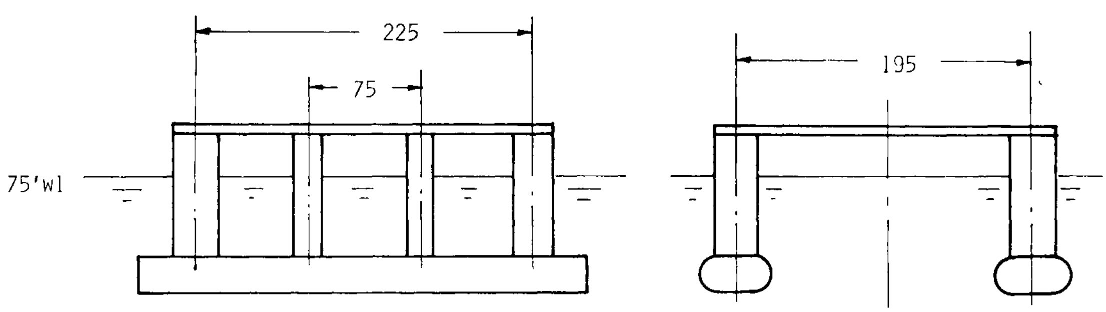
Three general procedures are in common use for the computation of the exciting forces and the coefficients in Equation (2). The first is three-dimensional potential flow theory applied to the entire platform. This is the most general method and yields the most complete results for situations in which viscous forces do not play an important role. Second, we may utilize a composite theory or synthesis involving the summation of forces computed for several simple geometric shapes into which the platform may be subdivided. These elementary solutions may be obtained by either potential theory solutions for elementary members such as horizontal or vertical circular cylinders, or by Morison's formula (Morison, et al, (1950)).

In the potential theory methods, the fluid forces are obtained by integrating a resultant pressure over the wetted surface of the platform and viscous shear forces are neglected. The process requires the solution for a velocity potential, $\phi(x,y,z,t)$, which satisfies the Laplace equation

$$\nabla^2\phi = 0 \tag{3}$$

in the fluid domain and which satisfies appropriate boundary conditions on the surface of the platform, on the free surface and at infinity. In the linear solution procedure, the potential function is expressed as the sum of three terms

$$\phi = \phi_I + \phi_D + \phi_M \tag{4}$$



Main hulls 21 x 50 x 300 ft.
Main columns 30 ft. dia.
Intermediate columns 15 ft. dia.
Weight 48 x 10^6 lbs.

Figure 1
Twin Hull Semisubmersible

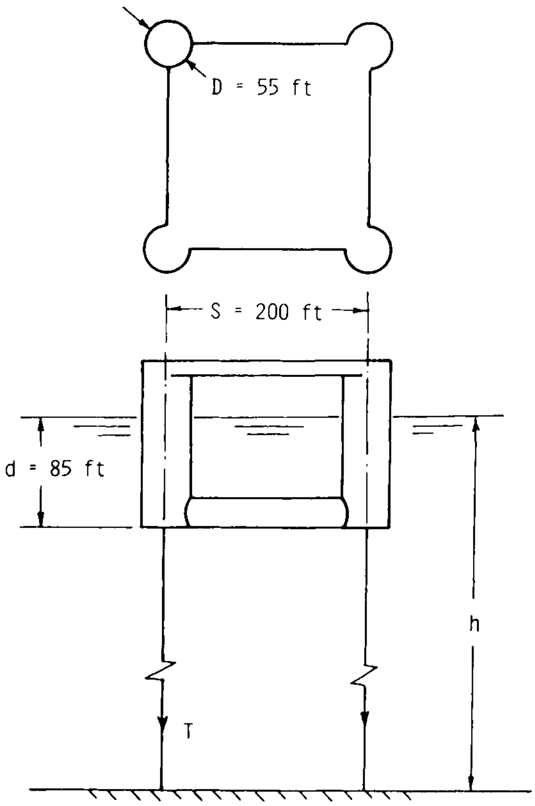


Figure 2
Four Column Tension Leg Platform

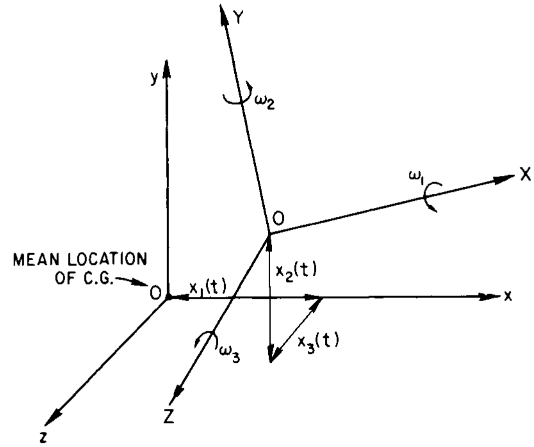


Figure 3
Coordinate Systems

Here, ϕ_I is the potential associated with the incident wave system, ϕ_D is the potential of a system of diffracted waves caused by the interaction of the platform with the incident waves, and ϕ_M is the potential which expresses the effect of the motion of the body. The pressure is given by the linearized Bernoulli equation

$$p = -\rho g y - \rho \frac{\partial \phi}{\partial t} \tag{5}$$

When the total force is evaluated, it is found that the first two terms of the potential function result in a time-varying exciting force which is independent of the platform motion. If sinusoidal motion is assumed, the third term, ϕ_M , results in two force components, one in phase with the body acceleration called the added mass and the other in phase with the velocity called the damping. As a result of this linear decomposition of the velocity potential, we see that the motion dependent forces are computed in the absence of incoming waves and the wave forces are computed as though the body were stationary.

The principal difficulty in applying the three-dimensional method to a practical offshore platform lies in finding the potential function which satisfies the necessary boundary conditions on the geometrically complex surface of the platform. The great advantage of the procedure is that it inherently contains the hydrodynamic interaction between the different members of the platform which may be lost in the other two methods. A number of different analytical or numerical procedures are used in obtaining the three-dimensional solutions. One in very common use is the distributed source procedure as published, for example by Garrison (1977), Faltinsen and Michelsen (1974) and Hogben and Standing (1974).

In some cases, regularity of the platform geometry may permit its being approximated by several individual members of simple geometry for which somewhat simpler potential solutions are available. Examples are the pontoon members of the platform which may be approximated by either a single cylinder or a pair of cylinders, and vertical cylindrical members. Solutions for the single horizontal cylinder have been given by Bolton and Ursell (1973). Solutions for slender horizontal members of arbitrary shape may be obtained by the Frank close fit method, Frank (1967) which has been extended by Lee (1971) to treat two parallel horizontal cylinders similar to the pontoons of a catamaran. This procedure has been embedded in a general platform motions program by Hong and Paulling (1977). Such solutions for horizontal cylinders may be combined to represent a member whose cross section varies along the length, resulting in the "strip theory" of ship motions in waves.

Solutions for a vertical circular cylinder are useful in treating the forces on the columns of the platform. Solutions for a vertical cylinder extending to the sea floor is given by McCamy and Fuchs (1954). Solutions for a cylinder of finite depth are given by Garrett (1971) and Eatock-Taylor (1979). A solution for a group of vertical cylinders is given by Ohkusu (1974).

The solutions for single horizontal or vertical cylinders would not contain the member-member interactions referred to above. The multi-cylinder solutions would contain such interactions, but in the strip theory solutions, this would be present only in the two-dimensional sense. In some cases, the interaction results in appreciable modification to the resultant combined fluid forces.

The Morison formula represents an intuitive attempt to include the viscous drag forces on the cylindrical member by means of a force proportional to the square of the relative velocity between fluid and cylinder. Such a viscous force is expected to be of importance in platforms having members of relatively small cross section. In this case, the spacing between members is often large compared to the member cross-sectional dimensions and the hydrodynamic disturbance introduced into the flow by the presence of other members may be neglected. The force on a single cylindrical member is then obtained by computing the fluid pressures, velocities and accelerations at the member location neglecting the presence of the other members. The flow properties are computed by an appropriate wave theory and the kinematics of the platform motion are added to express the motion relative to the member. A form of the Morison formula in terms of the relative motion is

$$\underline{F} = -\iint_s p\,\underline{n}\,ds + \int_l C_D \underline{u}_n |\underline{u}_n| dl + \int_l C_M \underline{a}_n dl \tag{6}$$

Here p = pressure which would exist in the undisturbed flow field at a point on the member surface.

$\underline{n}$ = unit outward normal vector.

C_D = quadratic drag coefficient for flow normal to the member centerline.

$\underline{u}_n$ = component of resultant velocity normal to the centerline of the member.

C_M = acceleration force coefficient.

$\underline{a}_n$ = component of resultant acceleration normal to the centerline of the member.

Note that both $\underline{u}_n$ and $\underline{a}_n$ are the resultant of the wave-induced fluid motion and the motion of the member which results from the rigid-body motion of the platform. The relative motion form of (6) is an intuitive extension of the formula originally proposed for computing the forces exerted by waves on a vertical stationary piling.

If the Morison formula is to be used in the linearized Equation (2), the quadratic drag term presents an obstacle. In this case, the quadratic term may be replaced by an equivalent linear drag force provided that the quadratic force is not inherently responsible for any effects such as sub- or superharmonic excitation which are not also contained in the linear form. In many but not all cases, this is a good assumption and we shall examine some of the exceptions later. The equivalent linear drag coefficient, C_{DL}, is defined as follows

$$C_{DL} \underline{u}_n = C_D \underline{u}_n \mid \underline{u}_n \mid \tag{7}$$

An optimal estimate of C_{DL} may be made by means of a procedure which minimizes the mean square error between the linear force and the "exact" quadratic force. For regular waves, the result is

$$C_{DL} = \frac{8u_o}{3\pi} C_D \tag{8}$$

where u_o = the amplitude of the sinusoidal relative velocity.

For random waves, the result is

$$C_{DL} = \sqrt{\frac{8}{\pi}}\, \bar{u}\, C_D \tag{9}$$

where $\bar{u}$ = RMS value of the randomly varying resultant normal velocity.

A similar procedure may be used as shown by Paulling (1980) in the case of a steady current superimposed on the oscillatory relative velocity. In this case, there is a steady mean force whose magnitude exceeds the drag force due to the current alone, and the equivalent oscillatory force exceeds the force which would result from the oscillatory component of velocity alone.

In all of these cases, the equivalent linear coefficient is found to depend on the magnitude of the relative velocity which, in turn, depends on both the wave-induced velocity and on the body motion. Since the latter is initially unknown and is to be found by solving Equation (2), an iterative solution procedure must be employed. Computational procedures based on the Morison formula have received wide acceptance in the offshore industry and are found to yield results in good agreement with experiments provided care is exercised in the choice of the coefficients of drag and added mass.

FIRST ORDER HEAVING RESPONSE OF SEMISUBMERSIBLES

We shall first examine in some detail the heave response of the semisubmersible platform consisting of vertical and horizontal members as depicted in Figure 1. A good approximation to the heave force exerted by waves of small amplitude on cylindrical members of small cross section is obtained from only the first and last terms of the Morison formula (6). Using this expression in conjunction with the pressure and acceleration obtained from linear infinitesimal wave theory, the vertical force exerted by beam seas on one of a pair of parallel horizontal circular cylinders will be given by

$$Y_H = -2\rho A_H L a \omega^2 e^{-kd} \cos(kb - \omega t) \tag{10}$$

Here A_H = cross sectional area of one horizontal cylinder.

L = length of cylinder.

$2b$ = horizontal separation of the two cylinders.

d = depth from mean water surface to centerline of cylinder.

The force on all of the vertical surface-piercing members on one side of the platform is given by

$$Y_V = \rho g\, A_V\, a\, e^{-kd} \cos (kb - \omega t) \tag{11}$$

Here A_V = total cross sectional area of all vertical members on one side.

It is assumed that the depth of the lower end of the vertical member is approximately equal to the depth of the centerline of the horizontal member and this is consistent with the assumption of small member diameter.

The total vertical force on all members comprising both sides of the platform is given by

$$\begin{aligned} Y &= 2\, \rho g\, a\, e^{-kd}\, (A_V - 2k\, L\, A_H) \cos kb \cos \omega t \\ &= F_o\, e^{-kd}\, (1 - kb\, A_o) \cos kb \cos \omega t \end{aligned} \tag{12}$$

Here $F_o = 2\, \rho g\, A_V\, a$

$A_o = 2\, \frac{L}{b} \frac{A_H}{A_V}$

$k = \frac{\omega^2}{g}$

The amplitude of this force is the product of three terms which vary with the wave frequency in such a manner that there is force cancellation at certain frequencies and reinforcement at others. The individual terms are illustrated in the upper part of Figure 4 and their product in the lower part of this figure. For practical platforms, the most important portion of this composite function lies in the region $0 \leq kb \leq \frac{3\pi}{2}$. The remaining portion of the range corresponds to short waves in which the assumption of small member diameter no longer is valid and in which the forces are found to be negligibly small.

A comparison of this simplified formulation with the more exact strip theory computations is given in Figure 5. The curves labelled "slender member" have been obtained by the complete Morison formula including linearized viscous drag and a modified added mass coefficient appropriate to the noncircular horizontal members. The curves labelled strip theory have been obtained by an ideal fluid theory which takes into consideration the hydrodynamic interaction between the horizontal members. Results are given here for the two pontoons alone (lower hulls) and for the composite structure containing lower hulls and columns. From a comparison of this figure and Figure 4, we may conclude that the heave force cancellation occurring at the value of $T = 25$ results from the phase difference in the forces on vertical and horizontal members while the cancellation at $T = 9$ results from the phase difference due to the transverse separation of the members. For the example platform, it is seen that the effect of finite member cross section and interference between members is quite small, although this is not the case for all twin hull semisubmersibles.

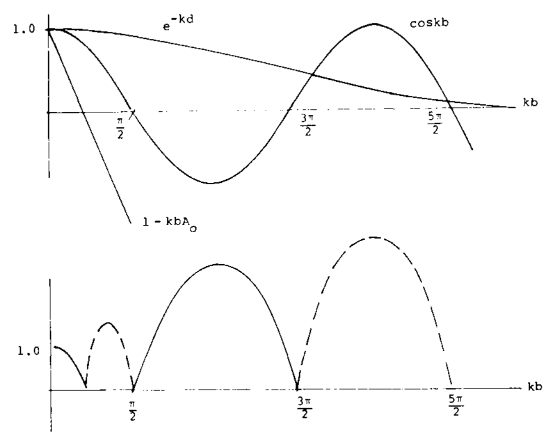


Figure 4
Heave Force on Twin Hull
Semisubmersible in Beam Seas

The vertical or heaving motion of the semisubmersible is usually decoupled from the other motions as a result of symmetry or near symmetry of the platform in the transverse and longitudinal directions. We may, therefore, consider heaving as a single degree of freedom motion with typical response characteristics of such a system. The restoration of this motion is primarily hydrostatic with a small contribution from the anchoring system in some cases, and the effective mass includes the added mass term in Equation (2). Figure 6 presents the added mass and damping for this platform as estimated by the Morison formula (slender member) and the strip theory. The equivalent linear damping coefficients are shown for the platform motion generated by three different sea states.

The platform natural frequency in heave is given by

$$\omega_n = \sqrt{\frac{c}{m + m'}} \tag{13}$$

In a typical platform the added mass is approximately equal to displaced mass of the pontoons, $2\,\rho A_H L$. The spring constant in heave, neglecting mooring effects is $2\,\rho g A_v$. Substituting in (13) gives the following expression for the natural

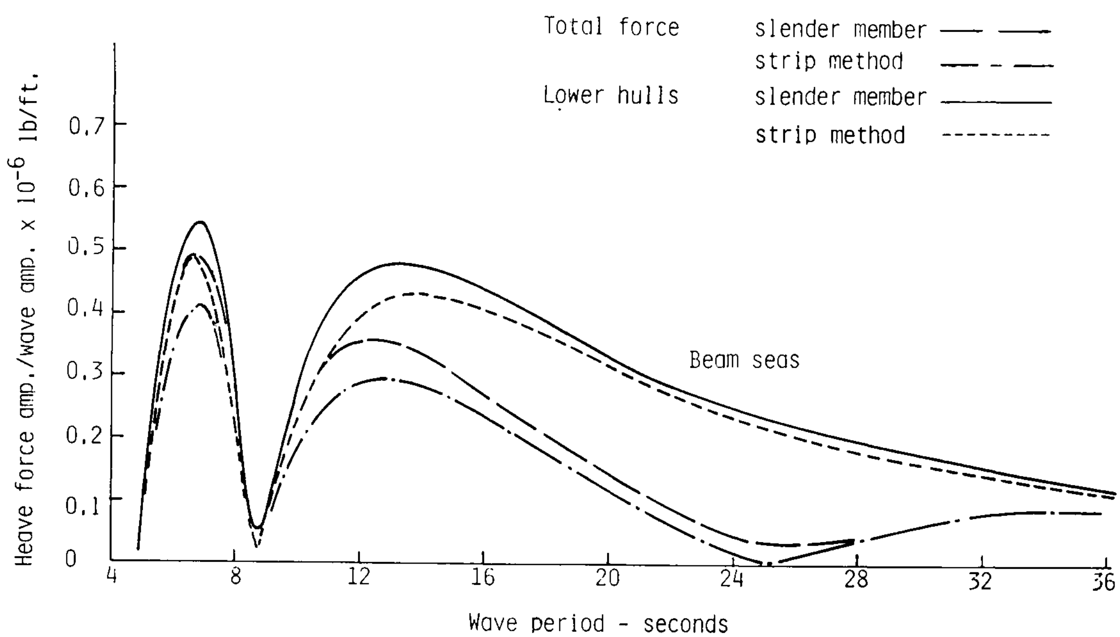


Figure 5
Twin Hull Semisubmersible Heave Exciting Force

frequency in heave.

$$\omega_n = \sqrt{\frac{2\rho g\, A_v}{2\rho(A_v\, d + 2\, A_H\, L)}} = \sqrt{\frac{g\,/\,d}{1 + 2\,\frac{V_H}{V_v}}} \tag{14}$$

Here V_v = the volume of vertical members.

V_H = the volume of horizontal members.

The resultant heave motion response to regular waves made nondimensional in terms of the wave amplitude is shown in Figure 7. The first minimum of the response to the left of the resonant peak corresponds to $kb = \pi/2$ and the second minimum to $kb = 1/A_o$. The remaining zeroes seen in Figure 4 are compressed into the left hand portion of the axis and, as noted, are of no practical importance.

The principles of force reinforcement and cancellation on members having different orientation in the platform structure plays an important role in the design of such platforms for optimum motion response in waves. By adjusting the ratios of horizontal to vertical surface-piercing member volume, the designer can adjust the frequencies of the response minima and control to a lesser extent the resonant peak. The amplitude of the latter peak and the minimum adjacent to it are strongly influenced by damping which may include important contributions from both viscous and radiation effects.

The typical ocean wave spectrum peak lies to the left of the resonance peak, and as a result, the mean response to waves is determined primarily by the peak just

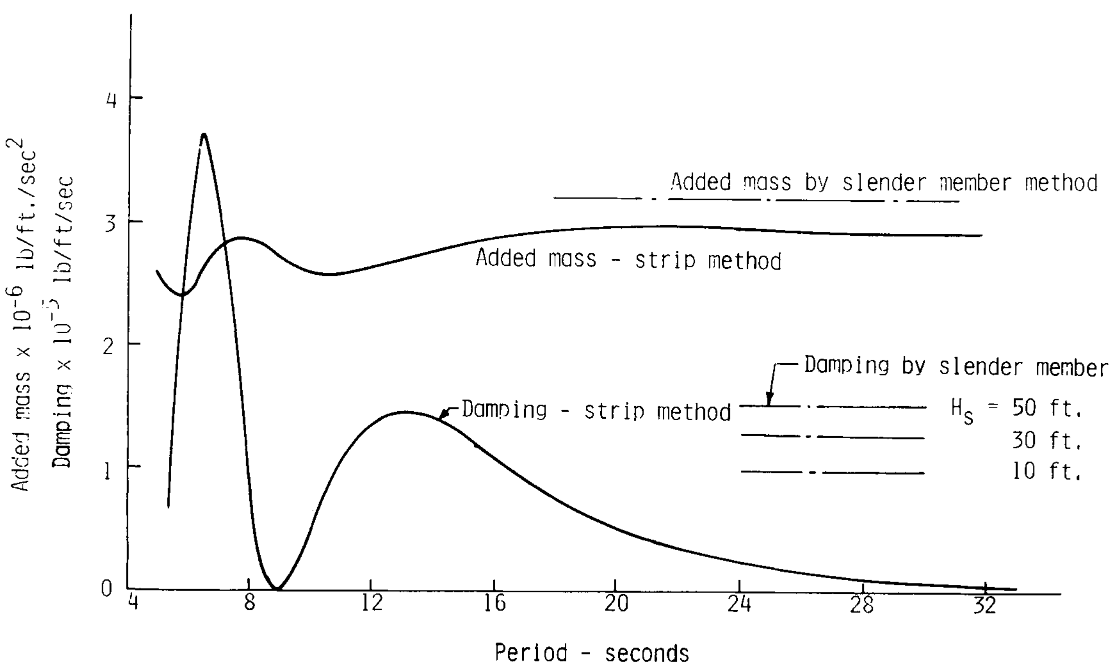


Figure 6
Twin Hull Semisubmersible - Heave Damping and Added Mass

below $kb = \pi/2$, together with the cancellation minimum at $kb = 1/A_o$. Common design practice is to choose the proportions of the platform such that the resonant frequency lies well outside the range of any appreciable wave energy.

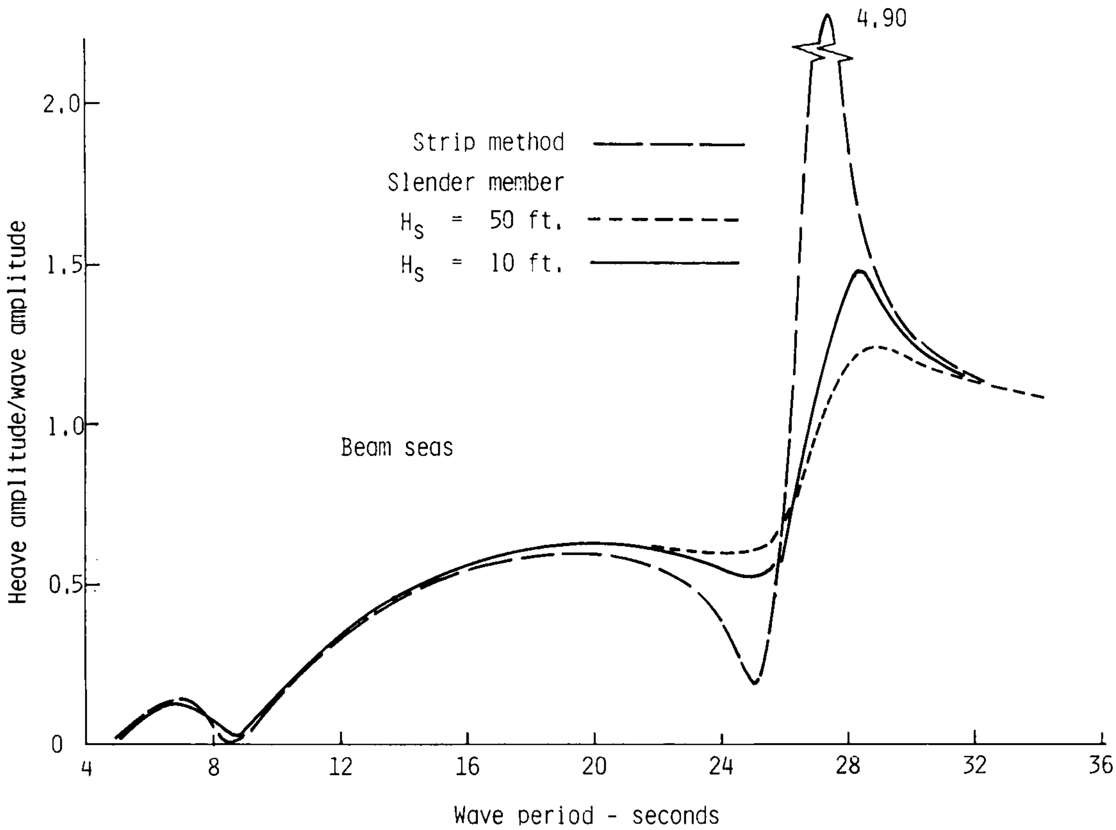


Figure 7

Twin Hull Semisubmersible Heave Response Function

RESPONSE OF TENSION LEG PLATFORMS

In contrast to the semisubmersible, the heave motion of the TLP is almost fully suppressed by the vertical mooring members. The lateral motions of surge, sway and yaw are only lightly restored by the vertical moorings. We may derive some important aspects of the TLP response by considering a simplified model somewhat similar to the previous semisubmersible space frame model. Figure 2 depicts a composite TLP similar to those which have been proposed for oil production in relatively deep water. An important design consideration in the TLP is the attainment of minimum wave-induced tension variation in the mooring members and it is found that this is obtained with relatively large vertical columns. The hydrodynamic interactions within groups of vertical cylinders, therefore, is important in TLP force computations. The horizontal (surge) exciting force is shown in Figure 8 as computed by a 3-D source distribution procedure and by the Morison formula. Two variations of the latter are shown. In the first, the fluid velocity and acceleration are computed on the member centerline as in the "pure" Morison formula. In the second, the average values of these parameters are computed over the member cross-section in order to provide an intuitive correction for the finite member dimensions.

Both the lateral motion (sway, surge) and the vertical (heave, roll) natural frequencies are of interest to the designer of the TLP. An estimate of the lateral natural frequency may be made as follows. From Figure 2 it is seen that the lateral restoring force coefficient is given by the net mooring tension divided by the length of the mooring lines. The tension must equal the difference between the platform buoyance and weight and, in deep water, the length of line is approximately equal to the water depth, h . The restoring coefficient is therefore given by

$$c = \frac{T}{h-d} = \frac{\Delta - w}{h} \qquad (15)$$

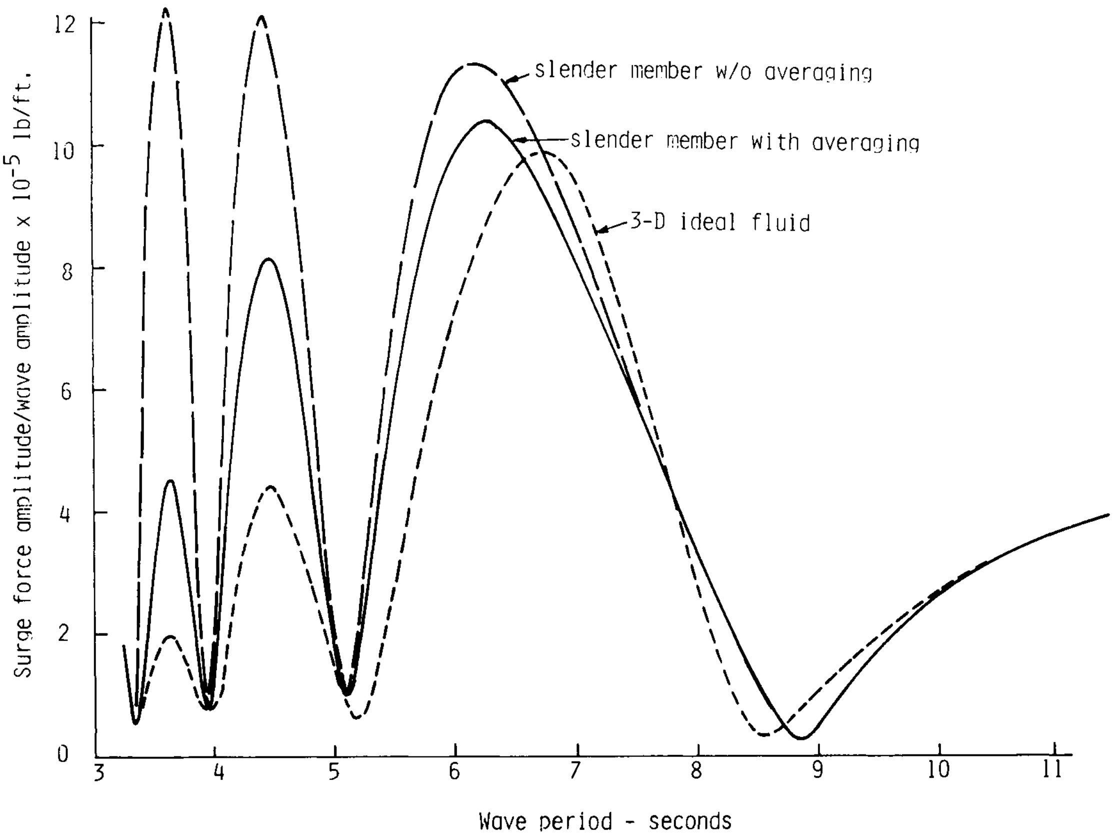


Figure 8
Four Column TLP Surge Force

For a typical TLP, the added mass in surge is approximately equal to the displaced water mass and the effective mass of the platform is therefore approximately equal to the sum of the physical mass and the displaced mass. Typical plots of the damping and added mass for the four column configuration are shown in Figures 9 and 10. Results are shown for the full four column configuration, a single column multiplied by four and for the frequency independent value given by the Morison formula. The effect of intermember interference is seen by a comparison of the former two values. The viscous damping by Morison's formula is seen to be small in comparison to the peak radiation damping in the range of short waves but is the only appreciable damping in the long wave range.

The natural frequency in sway or surge is given, approximately, by Equation (16)

$$\omega_n = \sqrt{\frac{cg}{\Delta + w}} = \sqrt{\frac{g}{h}\left(\frac{1 - w/\Delta}{1 + w/\Delta}\right)} \qquad (16)$$

In heave, the restoration coefficient is associated with the elastic stiffness of the mooring members and hydrostatic effects are neglible. Let A_m represents the total cross-sectional area of all mooring members, and, as before let us assume deep water so that h is the member length. The stiffness in heave is then given by

$$c = \frac{EA_m}{h} \tag{17}$$

where E is the material modulus of elasticity.

The added mass in heave is usually somewhat less than the displaced water mass since the principal part of the volume is in the vertical column members. As a reasonable approximation, the effective mass may be taken to be twice the physical mass for typical platform proportions, resulting in a natural frequency of heave given by Equation (18).

$$\omega_n = \sqrt{\frac{cg}{2w}} = \sqrt{\frac{E A_m g}{2 h w}} \tag{18}$$

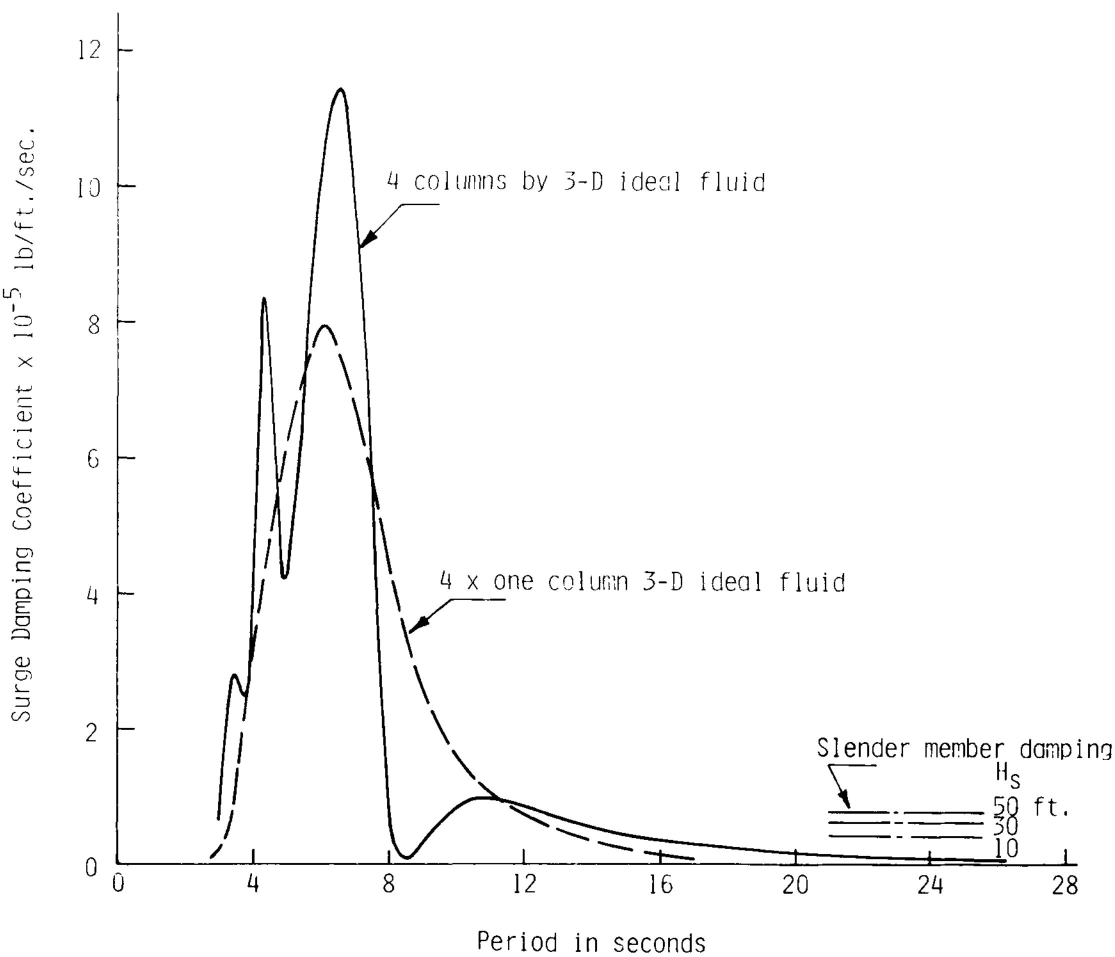


Figure 9
Four Column TLP Surge Damping

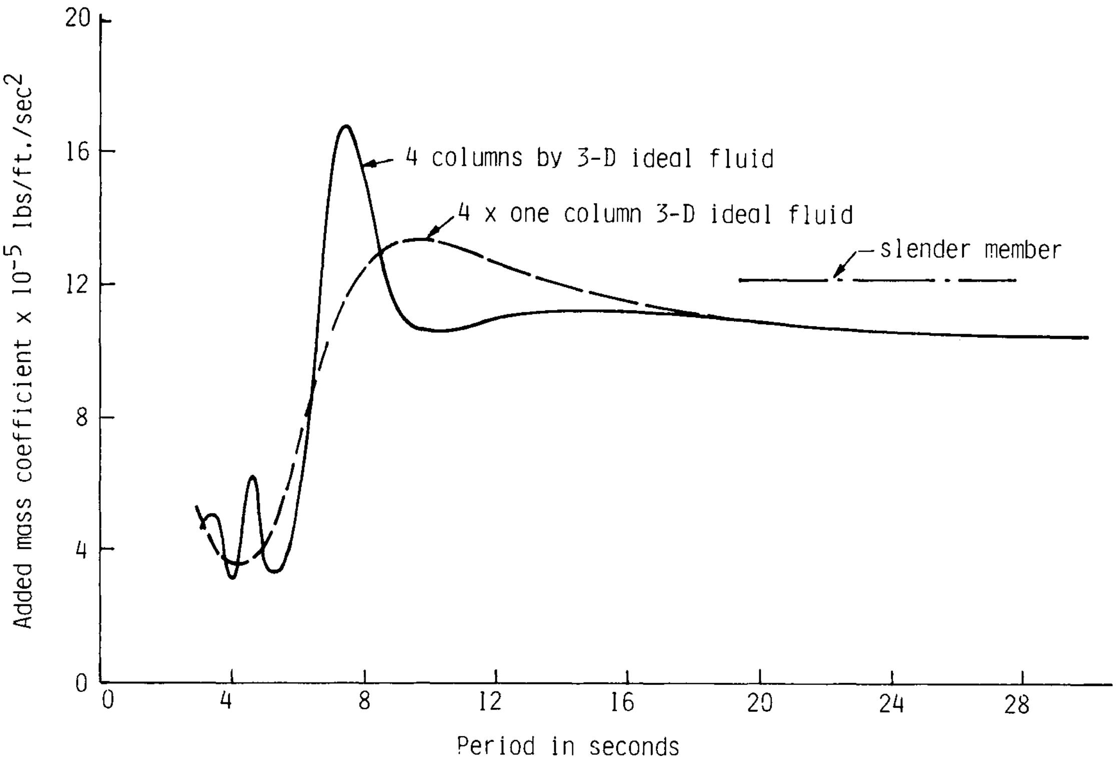


Figure 10
Four Column TLP Added Mass in Surge

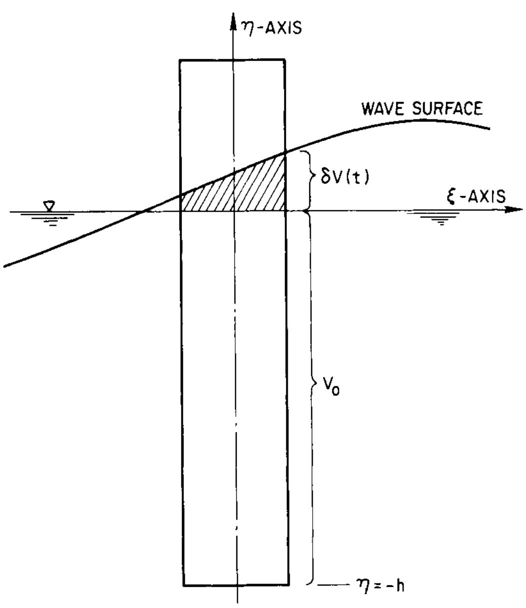


Figure 11
Surface-Piercing Vertical Cylinder

It is found that the frequency given by (16) is typical of all three lateral motions in surge, sway and yaw, while (18) is typical of the vertical plane motions of heave, roll and pitch. If we were to introduce the proportions of a typical (proposed) TLP for a water depth of 500 to 1000 meters, the natural periods of these categories of motions would be found to lie in the vicinity of 100 seconds and two to three seconds, respectively. These frequencies are outside the range of important wave frequencies and therefore, resonant amplification of the fundamental wave-induced response is usually of small concern. Since both the vertical and the lateral modes are lightly damped in the vicinity of their resonant frequencies, they can be troublesome due to excitation by nonlinear effects which result in forces which occur at either multiples or differences of important wave frequencies. Some of these effects will be discussed later and the accurate determination of the relevant forces forms one of the most important areas of needed research in the field.

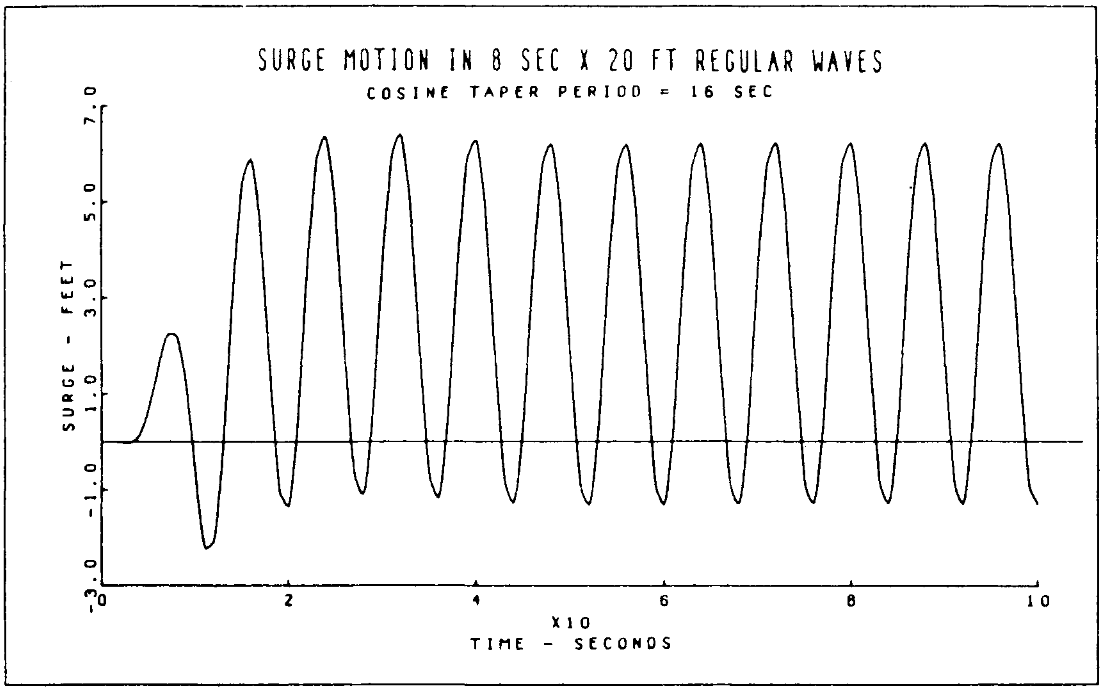


Figure 12
Surge Motion Time History for Experimental TLP

LOW AND HIGH FREQUENCY WAVE FORCES

Several unrelated fluid flow phenomena may contribute to wave-induced forces which have frequencies which differ by being integral multiples or fractions of the basic wave frequency. These second order-forces have been the subject of considerable recent investigation and details of their analyses may be found in papers by Pinkster (1979), Faltinsen (1978) and Newman (1967). Pinkster has computed the potential low frequency forces by integration of the second order pressure terms over the hull surface. It is shown that the most important contribution comes from the hydrostatic pressure integrated the region between the mean waterline and the instantaneous waterline. This effect is neglected in the usual linear analysis in which the integration is carried out only to the mean waterline.

A second source of such forces originates in the viscous drag on the alternately wet and exposed portion of the member projecting through the water surface. In Figure 11 we see a vertical surface-piercing cylindrical member exposed to wave action. When the member is in a wave crest, the drag force in the down wave direction acts on a wetted length of member extending up to the full height of the crest. In a wave trough, the force acts in the opposite direction but the wetted length on which the force acts is reduced by the wave height. In regular waves, the result consists of a steady force in the down wave direction plus an oscillatory force at twice the wave frequency. In random waves, the mean force becomes a slowly varying force having a frequency content similar to the wave envelope.

The characteristic high or low natural frequencies of the TLP as given by Equations (16) and (18) isolate the response from appreciable wave-frequency resonant amplification. The platform is, however, vulnerable to the high and low frequency effects described above. The high frequency effects are caused primarily by forces acting at the waterline and, as a result, these forces may exert an appreciable pitching moment on the platform. The resonant amplification of the lightly damped pitch motion, which has a natural frequency similar to that in heave, may result in severe oscillatory mooring member tensions with consequent accelerated fatigue damage in those members.

The low frequency and steady components of such forces may result in large offset of the platform from its mean position. Superimposed on this is the wave frequency oscillatory motion and effects caused by wind and current. In order for the platform to be capable of supporting drilling and other operations, this total offset must usually not exceed ten percent of the water depth. A time history of the motions of a small experimental TLP including the drift effect is shown in Figure 12. These results were obtained by a numerical integration of the complete equations of motion as described by Paulling (1977) and therefore include effects caused by large amplitude motions of the platform and waves.

CONCLUSIONS

The hydrodynamic forces on an offshore platform depend on both wave and platform motion and include both viscous and potential effects. The wave forces acting on different parts of the platform vary in phase and amplitude with wave frequency and, at certain frequencies, there may be either cancellation or reinforcement in their resultant. The platform response is strongly influenced by the relationship between these cancellation or reinforcement frequencies and the natural frequencies of the platform motions. The designer can sometimes choose the proportions of the platform in such a way as to take advantage of these force cancellation effects in order to achieve minimum motion response in a seaway.

In some cases, notably the tension leg platform, the natural frequencies are widely separated from the wave frequencies. Motion excitation may nevertheless occur as a result of second order low or high frequency effects. The accurate determination of these forces as well as the platform damping and other hydrodynamic characteristics in the range of very low and very high frequencies are important problem areas in which research is urgently needed.

REFERENCES

[1] Bolton, W. E. and Ursell, F., The Wave Force on an Infinitely Long Circular Cylinder in an Oblique Sea, J. Fluid Mech. (1973) v 57 pt 2, pp 241-256.

[2] Faltinsen, O. M. and Løken, A. E., Drift Forces and Slowly Varying Forces on Ships and Offshore Structures, Nor. Marit. Res., No. 1, (1978) pp 2-15.

[3] Faltinsen, O. M. and Michelsen, F., Motions of Large Structures in Waves at Zero Froude Number, Proc. Intl. Symp. on Dynamics of Marine Vehicles and Structures in Waves, Inst. Mech. Engrs., London, April 1974.

[4] Frank, W. , Oscillations of Cylinders in or Below the Free Surface of Deep Fluids, NSRDC Report 2375, Oct. 1967.

[5] Garrett, C. J. R., Wave Forces on a Circular Dock, J. Fluid Mech. (1971), v 46, pt 1, pp 129-139.

[6] Garrison, C. J., Hydrodynamic Interaction of Waves with a Large-Displacement Floating Body, U.S. Navy Postgraduate School, Report NPS-69Cm77091, Sept. 1977.

[7] Hammett, D. S., The First Dynamically Stationed Semisubmersible Sedco 709, Offshore Tech. Conf., Houston Paper No. 2972, 1977.

[8] Hogben, N. and Standing, R. G., Wave Loads on Large Bodies, Proc. Intl. Symp. on Dynamics of Marine Vehicles and Structures in Waves, Inst. Mech. Engrs., London, April 1974.

[9] Horton, E. E., McCammon, L. B., Murtha, J. P. and Paulling, J. R., Optimization of Stable Platform Characteristics, Offshore Tech. Conf., Houston, Paper No. 1553,1972.

[10] Lee, C. M., Jones, H. and Bedel, J. W., Added Mass and Damping Coefficients of Heaving Twin Cylinders in a Free Surface, NSRDC Report 3695, Aug. 1971.

[11] Lighthill, J., Waves and Hydrodynamic Loading, Proceedings BOSS 1979, pp 1-40.

[12] McCamy, R. C. and Fuchs, R. A., Wave Forces on Piles: A Diffraction Theory Beach Erosion Board, U.S. Army Corps of Engineers, Tech. Memo 69, 1954.

[13] Morison, J. R., O'Brien, M. P., Johnson, J. W. and Schaaf, S. A., The Force Exerted by Surface Waves on Piles, Pet. Trans., AIME, v 189, 1950.

[14] Newman, J. N., The Drift Force and Moment on Ships in Waves, J. Ship Research, 1967, pp 51-60.

[15] Ohkusu, M, On the Hydrodynamic Forces on Multiple Cylinders in Waves, Proc. Intl. Symp. on Dynamics of Marine Vehicles and Structures in Waves, University College London, 1974, pp 115-120.

[16] Paulling, J. R. and Horton, E. E., Analysis of the Tension Leg Platform, Offshore Tech. Conf., 1263, 1970.

[17] Paulling, J. R. and Hong, Y. S., Analysis of Semisubmersible Catamaran-type Platforms, Offshore Tech. Conf., Houston, Paper No. 2975, 1977.

[18] Paulling, J. R., Time-Domain Simulation of Semisubmersible Platform Motion with Application to the Tension-Leg Platform, SNAME Spring Meeting/STAR Symposium, San Francisco, 1977.

[19] Paulling, J. R., An Equivalent Linear Representation of the Forces Exerted on the OTEC CW Pipe by Combined Effects of Waves and Current, Ocean Energy for OTEC, OED, v 9, ASME 1980, pp 21-28.

[20] Paulling, J. R., The Sensitivity of Predicted Loads and Responses of Floating Platforms to Computational Methods, PROC. Conf. on Integrity of Offshore Structures, Glasgow, 1981, (Pub. by Applied Science Publishers, Essex, U. K.,) 1981.

[21] Pinkster, J. A., Mean and Low Frequency Wave Drifting Forces on Floating Structures, Ocean Engng., v 6, 1979, pp 593-616.

[22] Salvesen, N., et. al, Computations of Nonlinear Surge Motions of Tension Leg Platforms, Offshore Tech. Conf., 4394, 1982.

[23] Taylor, R. Eatock and Dolla, J. P., Hydrodynamic Loads on Vertical Bodies of Revolution, Trans. RINA, v 122, 1980, pp 285-298.